Lecture 23: The Dot Product

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

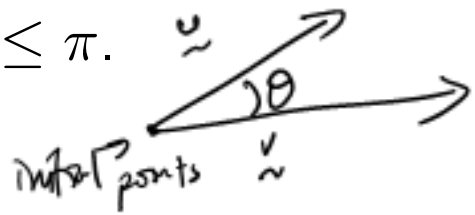
Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define and compute the dot product of two vectors in both two and three dimensions.
- ☐ Determine the angle between two vectors using the dot product.
- ☐ Verify whether two vectors are orthogonal.
- ☐ Apply properties of the dot product in vector algebra.
- ☐ Apply the dot product to solve problems involving projections of one vector onto another. $\text{proj}_{\underset{\sim}{v}} \underset{\sim}{u}$

Angle Between Vectors

When two nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are placed so their initial points coincide, they form an angle θ of measure $0 \leq \theta \leq \pi$.

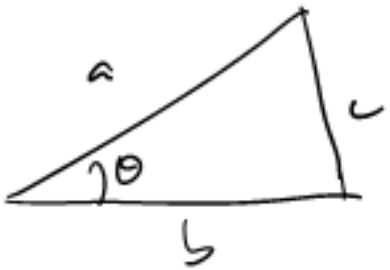


Angle Between Two Vectors

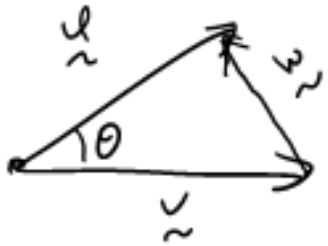
The angle θ between two nonzero vectors $\boldsymbol{u} = \langle u_1, u_2, u_3 \rangle$ and $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ is given by

$$\theta = \cos^{-1}\left(\frac{u_1 v_1 + u_2 v_2 + u_3 v_3}{|\boldsymbol{u}||\boldsymbol{v}|}\right).$$

Recall: Law of cosine (Calculus I)



$$c^2 = a^2 + b^2 - 2ab\cos\theta$$



$$\underset{\sim}{u} = \underset{\sim}{v} + \underset{\sim}{w}$$

$$\underset{\sim}{w} = \underset{\sim}{u} - \underset{\sim}{v}$$

Law of cosine : $|\underset{\sim}{w}|^2 = |\underset{\sim}{u}|^2 + |\underset{\sim}{v}|^2 - 2|\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta$

$$2|\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta = |\underset{\sim}{u}|^2 + |\underset{\sim}{v}|^2 - |\underset{\sim}{w}|^2$$

$$= \left| \langle u_1, u_2, u_3 \rangle \right|^2 + \left| \langle v_1, v_2, v_3 \rangle \right|^2$$

$$- \left| \langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle \right|^2$$

$\underset{\sim}{w} = \underset{\sim}{u} - \underset{\sim}{v}$

$$= \left(\sqrt{u_1^2 + u_2^2 + u_3^2} \right)^2 + \left(\sqrt{v_1^2 + v_2^2 + v_3^2} \right)^2$$

$$- \left(\sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2} \right)^2$$

$$2|\underset{\sim}{u}||\underset{\sim}{v}|\cos\theta = u_1^2+u_2^2+u_3^2 + v_1^2+v_2^2+v_3^2$$
$$- \left(u_1^2 - 2u_1v_1 + v_1^2 + u_2^2 - 2u_2v_2 + v_2^2 \right.$$
$$\left. + u_3^2 - 2u_3v_3 + v_3^2 \right)$$
$$= 2u_1v_1 + 2u_2v_2 + 2u_3v_3$$
$$\cos\theta = \frac{u_1v_1 + u_2v_2 + u_3v_3}{|\underset{\sim}{u}||\underset{\sim}{v}|}$$
$$\theta = \arccos\left(\frac{u_1v_1 + u_2v_2 + u_3v_3}{|\underset{\sim}{u}||\underset{\sim}{v}|}\right) \leftarrow \underset{\sim}{u}\cdot\underset{\sim}{v}$$

Dot Product

Given two vectors $\boldsymbol{u} = \langle u_1, u_2, u_3 \rangle$ and $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$, the dot product ($\boldsymbol{u} \cdot \boldsymbol{v}$) is the scalar

scalar product
inner product.

$$\boldsymbol{u} \cdot \boldsymbol{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

← number / scalar.

Example:

(a) $\langle 1, -2, -1 \rangle \cdot \langle -6, 2, -3 \rangle = (1)(-6) + (-2)(2) + (-1)(-3)$

$= -6 - 4 + 3 = -7$ ← a number

(b) $\left(\frac{1}{2}\boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}\right) \cdot (4\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k}) = \left(\frac{1}{2}\right)(4) + (3)(-1) + (1)(2)$

$= 2 - 3 + 2 = 1$

Dot Product

Dot Product and Angle

The angle of two nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is $\theta = \cos^{-1}\left(\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{u}||\boldsymbol{v}|}\right)$.

The dot product of two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is given by $\boldsymbol{u}\cdot\boldsymbol{v} = |\boldsymbol{u}||\boldsymbol{v}|\cos\theta$.

Example: Find the angle between $\boldsymbol{u} = \boldsymbol{i} - 2\boldsymbol{j} - 2\boldsymbol{k}$ and $\boldsymbol{v} = 6\boldsymbol{i} + 3\boldsymbol{j} + 2\boldsymbol{k}$.

$$\underset{\sim}{u}\cdot\underset{\sim}{v} = (1)(6)+(-2)(3)+(-2)(2) = -4$$

$$|\underset{\sim}{u}| = \sqrt{(1)^2+(-2)^2+(-2)^2} = 3$$

$$|\underset{\sim}{v}| = \sqrt{6^2+3^2+2^2} = 7$$

$$\theta = \arccos\left(\frac{\underset{\sim}{u}\cdot\underset{\sim}{v}}{|\underset{\sim}{u}||\underset{\sim}{v}|}\right) = \arccos\left(\frac{-4}{3\cdot 7}\right) = \arccos\left(\frac{-4}{21}\right)$$

$$\approx 1.76 \text{ rad.}$$

Orthogonal Vectors

Two non-zero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are orthogonal (or perpendicular) if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$.

$\underset{\sim}{u} \cdot \underset{\sim}{v} = 0 \Rightarrow \underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta = 0$

If $|\underset{\sim}{u}| \neq 0$, $|\underset{\sim}{v}| \neq 0$, then $\cos\theta = 0 \Rightarrow \theta = \pi/2$

Example: Determine if the following vectors are orthogonal.

(a) $\boldsymbol{u} = \langle 3, -2 \rangle$, $\boldsymbol{v} = \langle 4, 6 \rangle$

$\underset{\sim}{u} \cdot \underset{\sim}{v} = (3)(4) + (-2)(6) = 0$

So $\underset{\sim}{u} \perp \underset{\sim}{v}$

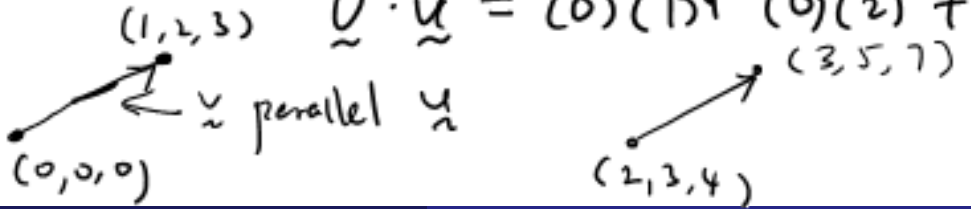
(b) $\boldsymbol{u} = 3\boldsymbol{i} - 2\boldsymbol{j} + \boldsymbol{k}$, $\boldsymbol{v} = 2\boldsymbol{j} + 4\boldsymbol{k}$

$\underset{\sim}{u} \cdot \underset{\sim}{v} = (3)(0) + (-2)(2) + (1)(4) = 0$

So $\underset{\sim}{u} \perp \underset{\sim}{v}$

(c) $\mathbf{0}$, $\boldsymbol{u} = \langle 1, 2, 3 \rangle$

$\underset{\sim}{0} \cdot \underset{\sim}{u} = (0)(1) + (0)(2) + (0)(3) = 0$



Dot Product Properties

If $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$ are any vectors and c is a scalar, then

1. $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$
2. $(c\boldsymbol{u}) \cdot \boldsymbol{v} = \boldsymbol{u} \cdot (c\boldsymbol{v}) = c(\boldsymbol{u} \cdot \boldsymbol{v})$
3. $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$
4. $\boldsymbol{u} \cdot \boldsymbol{u} = |\boldsymbol{u}|^2$
5. $\mathbf{0} \cdot \boldsymbol{u} = 0$

All of these can be easily verified by letting Exercise.

$$\underset{\sim}{u} = \langle u_1, u_2, u_3 \rangle, \quad \underset{\sim}{v} = \langle v_1, v_2, v_3 \rangle, \quad \underset{\sim}{w} = \langle w_1, w_2, w_3 \rangle$$

(4) $\underset{\sim}{u} \cdot \underset{\sim}{u} = \langle u_1, u_2, u_3 \rangle \cdot \langle u_1, u_2, u_3 \rangle$

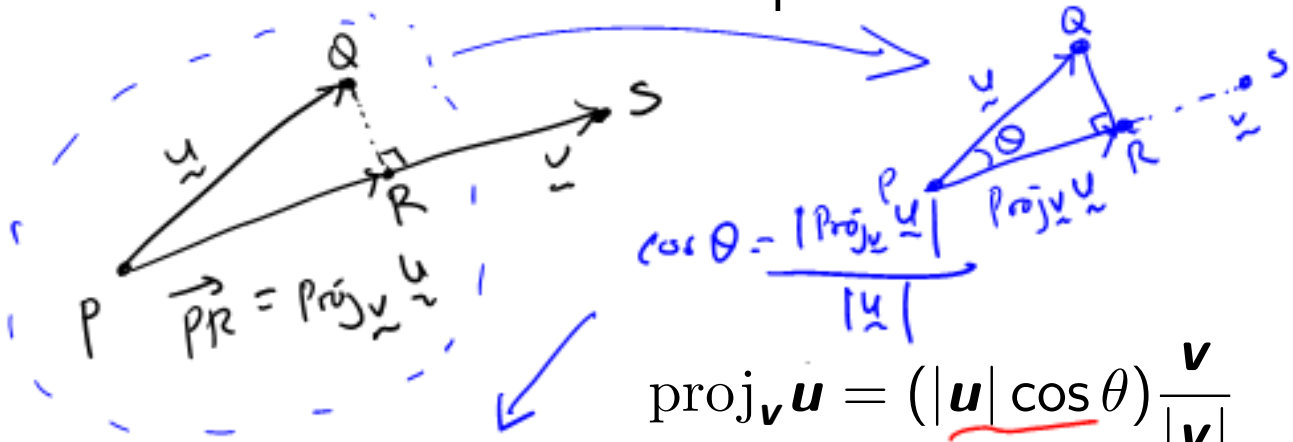
$= u_1^2 + u_2^2 + u_3^2 = \left(\sqrt{u_1^2 + u_2^2 + u_3^2}\right)^2 = |\underset{\sim}{u}|^2$

$|\underset{\sim}{u}|$

Vector Projections

The vector projection of $\boldsymbol{u} = \overrightarrow{PQ}$ onto a nonzero vector $\boldsymbol{v} = \overrightarrow{PS}$ is the vector $\overrightarrow{PR}$ determined by dropping a perpendicular from Q to the line PS, denoted by $\mathrm{proj}_{\boldsymbol{v}}\boldsymbol{u}$ (read as the vector projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$).

You may think this as the 'shadow' of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$. It is also sometimes referred as the component of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$.



we want to find a vector with the magnitude $|\mathrm{Proj}_{\underset{\sim}{v}}\underset{\sim}{u}|$ in the direction of $\underset{\sim}{v}$

$$\mathrm{proj}_{\boldsymbol{v}}\boldsymbol{u} = (|\boldsymbol{u}|\cos\theta)\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$$

length ← $|\boldsymbol{u}|\cos\theta$; direction (unit vector along the direction) ← $\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$

$$= \left(\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|}\right)\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$$

scalar component ← $\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|}$; direction component ← $\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$

$$= \left(\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|^2}\right)\boldsymbol{v}.$$

$|\mathrm{Proj}_{\underset{\sim}{v}}\underset{\sim}{u}| = |u|\cos\theta$

$\underset{\sim}{u}\cdot\underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}|\cos\theta$

$|\underset{\sim}{u}|\cos\theta = \frac{\underset{\sim}{u}\cdot\underset{\sim}{v}}{|\underset{\sim}{v}|}$

Vector Projections

Example: Find the vector projection of $\boldsymbol{u} = 6\boldsymbol{i} + 3\boldsymbol{j} + 2\boldsymbol{k}$ onto $\boldsymbol{v} = \boldsymbol{i} - 2\boldsymbol{j} - 2\boldsymbol{k}$ and the scalar component of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$.

$$\text{Proj}_{\underset{\sim}{v}} \underset{\sim}{u} = \left(\frac{\underset{\sim}{u} \cdot \underset{\sim}{v}}{|\underset{\sim}{v}|^2} \right) \underset{\sim}{v}$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = (6)(1) + (3)(-2) + 2(-2) = -4$$

$$|\underset{\sim}{v}| = \sqrt{(1)^2 + (-2)^2 + (-2)^2} = 3$$

Scalar component:

$$\left(\frac{\underset{\sim}{u} \cdot \underset{\sim}{v}}{|\underset{\sim}{v}|} \right) = \frac{-4}{3}$$

$$\text{Proj}_{\underset{\sim}{v}} \underset{\sim}{u} = \left(\frac{-4}{3^2} \right) (\underset{\sim}{i} - 2\underset{\sim}{j} - 2\underset{\sim}{k})$$

$$= \frac{-4}{9} (\underset{\sim}{i} - 2\underset{\sim}{j} - 2\underset{\sim}{k})$$

$$= \frac{-4}{9}\underset{\sim}{i} + \frac{8}{9}\underset{\sim}{j} + \frac{8}{9}\underset{\sim}{k}$$

Vector Projections

Example: Find the vector projection of $\boldsymbol{u} = 5\boldsymbol{i} + 2\boldsymbol{j}$ onto $\boldsymbol{v} = \boldsymbol{i} - 3\boldsymbol{j}$ and the scalar component of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$.

Exercise

Answer : $\text{proj}_{\underset{\sim}{v}} \underset{\sim}{u} = \left\langle -\frac{1}{10}, \frac{3}{10} \right\rangle$

scalar component : $\frac{-1}{\sqrt{10}}$

Vector Projections

Example: Show that the vector $\boldsymbol{u} - \text{proj}_{\boldsymbol{v}}\boldsymbol{u}$ is orthogonal to the projection vector $\text{proj}_{\boldsymbol{v}}\boldsymbol{u}$.

Note that $\text{proj}_{\underset{\sim}{v}}\underset{\sim}{u}$ is parallel to the vector $\underset{\sim}{v}$. So it is enough to check that $\underset{\sim}{u} - \text{proj}_{\underset{\sim}{v}}\underset{\sim}{u}$ is orthogonal to $\underset{\sim}{v}$

$$\left(\underset{\sim}{u} - \text{proj}_{\underset{\sim}{v}}\underset{\sim}{u}\right)\cdot\underset{\sim}{v} = \underset{\sim}{u}\cdot\underset{\sim}{v} - \text{proj}_{\underset{\sim}{v}}\underset{\sim}{u}\cdot\underset{\sim}{v}$$

$$= \underset{\sim}{u}\cdot\underset{\sim}{v} - \left[\left(\frac{\underset{\sim}{u}\cdot\underset{\sim}{v}}{|\underset{\sim}{v}|^2}\right)\underset{\sim}{v}\right]\cdot\underset{\sim}{v}$$

scalar

$$= \underset{\sim}{u}\cdot\underset{\sim}{v} - \left(\frac{\underset{\sim}{u}\cdot\underset{\sim}{v}}{|\underset{\sim}{v}|^2}\right)(\underset{\sim}{v}\cdot\underset{\sim}{v})$$

$$= \underset{\sim}{u}\cdot\underset{\sim}{v} - \left(\frac{\underset{\sim}{u}\cdot\underset{\sim}{v}}{|\underset{\sim}{v}|^2}\right)|\underset{\sim}{v}|^2$$

$$= \underset{\sim}{u} \cdot \underset{\sim}{v} - \underset{\sim}{u} \cdot \underset{\sim}{v} = 0$$

So, $\underset{\sim}{u} - \text{proj}_{\underset{\sim}{v}} \underset{\sim}{u} \perp \underset{\sim}{v}$ and hence $\underset{\sim}{u} - \text{proj}_{\underset{\sim}{v}} \underset{\sim}{u} \perp \text{proj}_{\underset{\sim}{v}} \underset{\sim}{u}$

Now, notice that $\underset{\sim}{u} = \text{proj}_{\underset{\sim}{v}} \underset{\sim}{u} + \left(\underset{\sim}{u} - \text{proj}_{\underset{\sim}{v}} \underset{\sim}{u} \right)$

$$= \left(\frac{\underset{\sim}{u} \cdot \underset{\sim}{v}}{|\underset{\sim}{v}|^2} \right) \underset{\sim}{v} + \left(\underset{\sim}{u} - \frac{\underset{\sim}{u} \cdot \underset{\sim}{v}}{|\underset{\sim}{v}|^2} \underset{\sim}{v} \right)$$

parallel to $\underset{\sim}{v}$

orthogonal to $\underset{\sim}{v}$

* We can always express a non-zero vector $\underset{\sim}{u}$ as the sum of two orthogonal vectors.